

Nios[®] V/m Processor PIO LED Toggle Design

Agilex[™] 5 FPGA

Last updated: **2024-09-30**

Contents

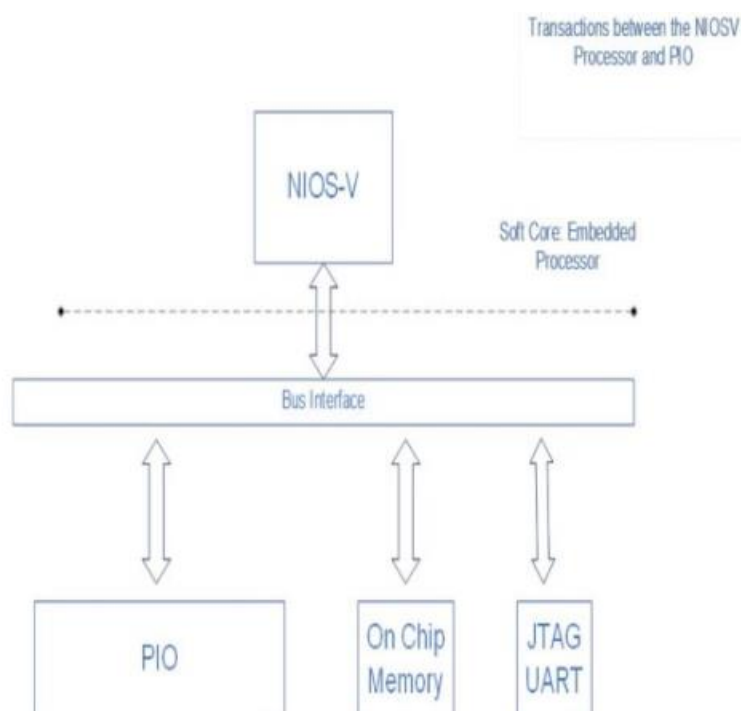
1.0	Theory of Operation	3
1.1	IP Cores	3
2.0	Simulating the design	4
3.0	Executing the design on Development Kit.....	5
3.1	Creating and validating the design	5
4.0	Expected Results	6
5.0	Document Revision History	9

1.0 Theory of Operation

This design demonstrates the transaction between the Nios® V processor and the Parallel Input/Output (PIO) core for the Agilex™ 5 FPGA E-Series 065B Premium Development Kit (ES1) DK-A5E065BB32AES1.

The PIO core is configured for output ports only and the outputs are connected to the LED on the development kit. The application, which runs atop this design, toggles these output registers of the PIO core. The application writes and reads back the content from the IP location.

Figure 1-1. Block Diagram



1.1 IP Cores

The following IPs are used in this design.

- Nios V/m soft processor core
- On Chip RAM-II
- JTAG UART
- PIO IP
- System ID

2.0 Simulating the design

Note: Please refer to the README.md file in the extracted design package from Quartus or in the sources folder of the Github repository of the design for the steps to create the design, application and generate the programming files.

- Unpackage/extract the design in your working directory.
- Simulate the design using Quartus Prime Pro tool.

3.0 Executing the design on Development Kit

3.1 Creating and validating the design

Note: Please refer to the README.md file in the extracted design package from Quartus or in the sources folder of the Github repository of the design for the steps to create the design, application and generate the programming files.

- Unpackage/extract the design in your working directory.
Locate the "ready_to_test" folder within the package.
- The folder contains the necessary files for executing the application on the board.
Refer to the readme file for the steps to program the application files on the board.
- Validate the design by observing the prints on the terminal.

4.0 Expected Results

The following is the output as observed on the JTAG UART terminal. The output is analogous to the logic from the application code. Users should be able to observe same output on their terminal/setup.

```
# Application to toggle the PIOs- [4:0]
# DATA READBACK FROM PIO_0_BASE is 0x0
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x1
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x2
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x3
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x4
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x5
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x6
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x7
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x8
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x9
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xa
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xb
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xc
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xd
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xe
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xf
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x0
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x1
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x2
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x3
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x4
```

4.0 Expected Results

```
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x5
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x6
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x7
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x8
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x9
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xa
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xb
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xc
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xd
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xe
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xf
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x0
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x1
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x2
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x3
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x4
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x5
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x6
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x7
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x8
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x9
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xa
```

4.0. Expected Results

```
# DATA READBACK FROM PIO_0_BASE is 0xb
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xc
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xd
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xe
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xf
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x0
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x1
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x2
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x3
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x4
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x5
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x6
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x7
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x8
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0x9
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xa
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xb
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xc
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xd
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xe
# DATA MATCHED - TEST PASSED
# DATA READBACK FROM PIO_0_BASE is 0xf
# DATA MATCHED - TEST PASSED
# NIOSV-PIO Test PASSED
```


5.0 Document Revision History

Date	Version	Changes
2024-09-30	24.3.0	Initial release on github

© Altera Corporation. Altera, the Altera logo, the 'a' logo, and other Altera marks are trademarks of Altera Corporation. Altera and Intel warrant performance of its FPGA and semiconductor products to current specifications in accordance with Altera's or Intel's standard warranty as applicable but reserves the right to make changes to any products and services at any time without notice. Altera and Intel assume no responsibility or liability arising out of the application or use of any information, product, or service described herein except as expressly agreed to in writing by Altera or Intel. Altera and Intel customers are advised to obtain the latest version of device specifications before relying on any published information and before placing orders for products or services.

*Other names and brands may be claimed as the property of others.